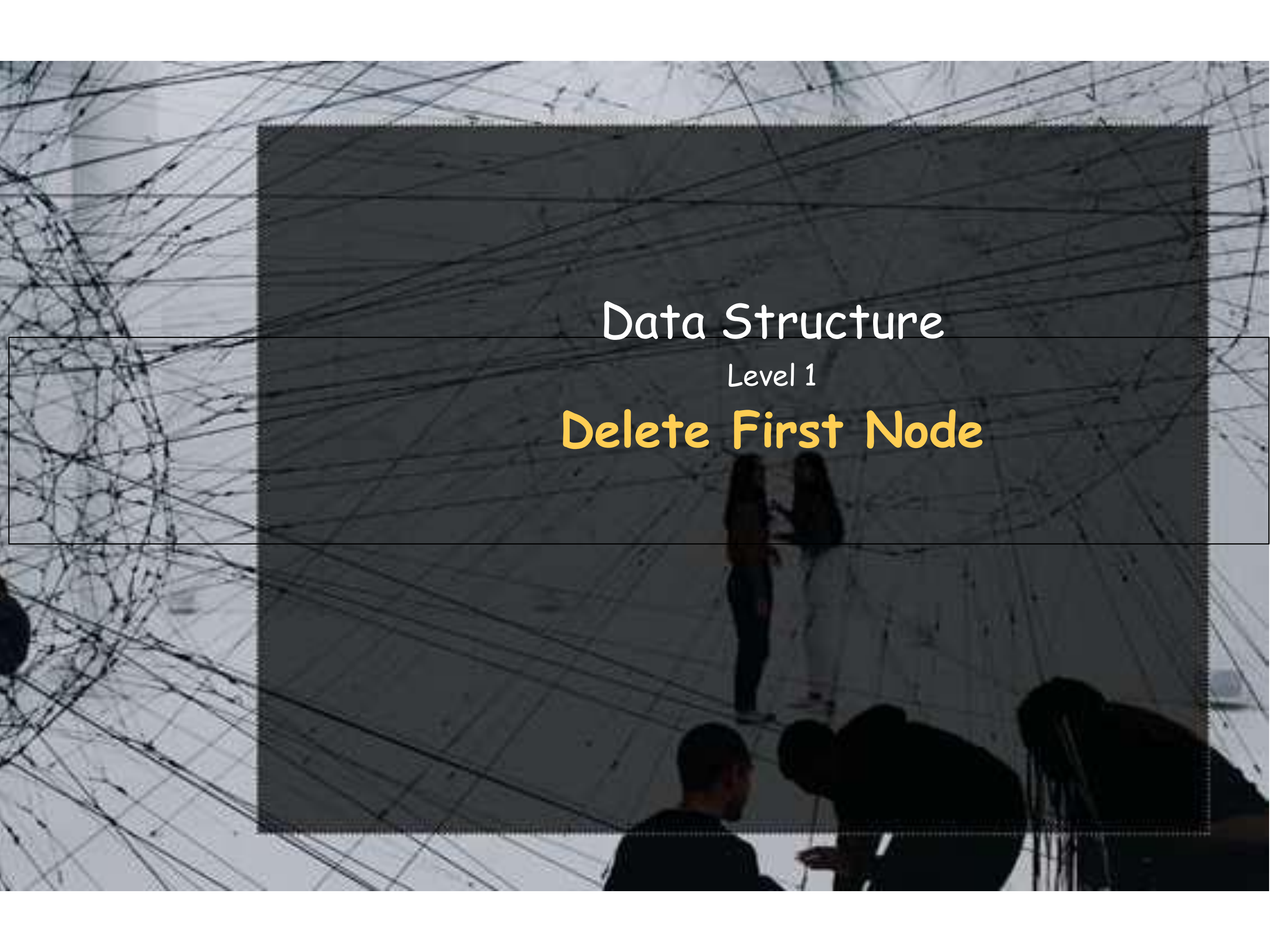


Data Structure

Level 1

Singly Linked List Data Structures Operations

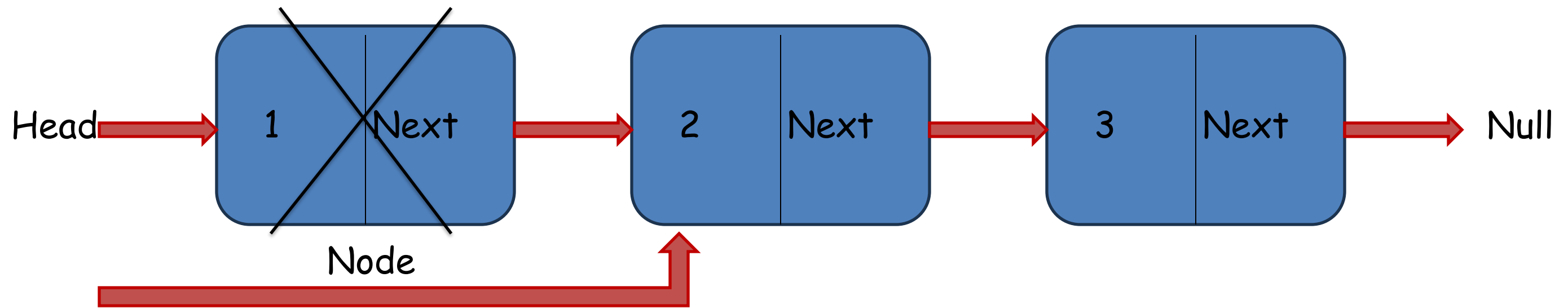
The background of the slide features a complex, abstract network of thin, dark lines crisscrossing a light gray background, resembling a neural network or a web. In the lower portion of the image, there are dark silhouettes of several people, some standing and some sitting, looking towards the center, suggesting a collaborative learning or meeting environment.

Data Structure

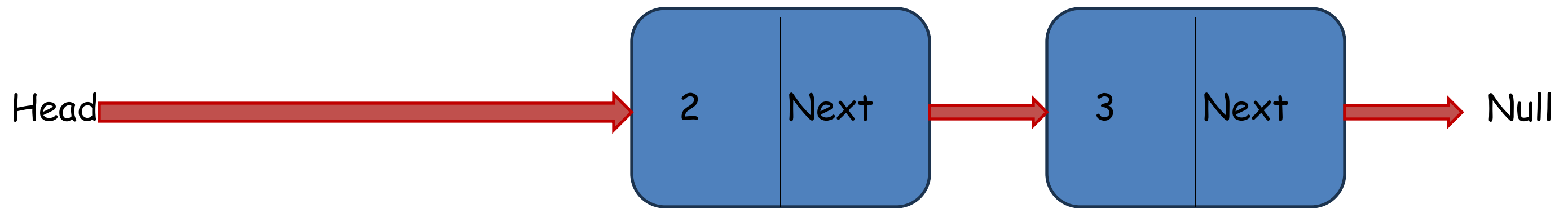
Level 1

Delete First Node

Linked List Data Structure



Linked List Data Structure





Definition

DeleteFirstNode() Function:

Removes the first node of the linked list and updates the head pointer.

Steps:

- Store the current head in a temporary pointer
- Move head to the next node
- Free memory of the old head



Explanation

- ◆ The head always points to the first node.
- ◆ When deleting the first node:
 - We shift the head pointer forward
 - The old first node is removed from memory
- ◆ No traversal is required.

Procedures (Delete First Node)

- 1 Check if the list is empty
- 2 Store Current = head
- 3 Move head $\rightarrow$ head = Current $\rightarrow$ next
- 4 Delete the old node $\rightarrow$ delete Current
- 5 Print updated list



Example

Initial list after insertion:

1 → 2 → 3 → 4 → 5 → 6 → NULL

After calling DeleteFirstNode(head)

➡ Resulting list:

2 → 3 → 4 → 5 → 6 → NULL

Note

This operation works only if head is properly passed by reference (Node*& head).



Comparison

- Delete First Node
 - ✓ Time Complexity: $O(1)$
 - ✓ No traversal needed
 - ✓ Simple pointer update

- Delete Specific Node
 - ✓ Requires search
 - ✓ Time Complexity: $O(n)$
 - ✓ Uses two pointers (Prev & Current)

Warning

If you forget to free memory using delete, a memory leak will occur.
Always check if head == NULL before deleting.

Summary

- ✓ DeleteFirstNode removes the head node
- ✓ Moves head to the next node
- ✓ Frees memory using delete
- ✓ Time Complexity is $O(1)$
- ✓ One of the simplest linked list operations

Thank you 😊